

Activity 1

Drag the words into the correct boxes

1. **Data** ✓ Facts, text, graphics, images, etc.
2. **Database application** ✓ Application program(s).
3. **Constraint** ✓ A rule that cannot be violated by database users.
4. **Repository** ✓ Centralized storehouse for all data definitions.
5. **Metadata** ✓ includes data definitions and constraints.
6. **Data warehouse** ✓ is an integrated decision support database.
7. **Information** ✓ Data placed in context or summarized.
8. **User view** ✓ A logical description of portion of database.
10. **Database managemen...** ✓ A software application that is used to create, maintain, and provide controlled access to user databases.
11. **Data independence** ✓ Separation of data description from programs.
12. **Database** ✓ Organized collection of related data.
13. **Enterprise resourc...** ✓ A business management system that integrates all functions of the enterprise.
14. **Systems developme...** ✓ A structured, step-by-step approach to systems development.
15. **Prototyping** ✓ A rapid approach to systems development.
16. **Enterprise data m...** ✓ A graphical model that shows the high-level entities for the organization and the relationships among those entities.
17. **Conceptual schema** ✓ A comprehensive description of business data.
18. **Internal schema** ✓ consists of two data models: a logical model and a physical model.
19. **External schema** ✓ consist of the enterprise data model and multiple user views

Flowchart

Logical schema

Activity 2

- a. data dependence; data independence
 - data dependence: all programs maintain metadata for each file they use; if the file structure changes, all related programs must also be modified.
 - data independence: metadata is handled by the DBMS, allowing changes to the database schema or storage structure without requiring changes in application programs
- b. structured data; unstructured data

data	
stored representations of meaningful objects and events	
structured data	unstructured data
number, text, date	image, video, document

- c. data; information
 - data: stored representations of meaningful objects and events
 - information: data processed to increase knowledge of the person using the data
- d. repository; database
 - repository: centralized storehouse of metadata
 - database: storehouse of the data; organized collection of logically related data
- e. data warehouse; ERP system

- erp: integrate all enterprise functions including manufacturing, finance, sales, marketing inventory, accounting, human resources
- data warehouse: integrated decision support system derived from various operational databases
- f. systems development life cycle; prototyping
 - sdlc: detailed, well-planned development process; time-consuming but comprehensive; long development cycle
 - prototyping: rapid application development (rad); cursory attempt at conceptual data modeling; define database during development of initial prototype; repeat implementation and maintenance activities with new prototype versions
- g. enterprise data model; project data model
 - enterprise data model: high-level entities and relationships for the organization; (first step in the database development process; specifies scope and general content; overall picture of organizational data at high level of abstraction; erd; descriptions of entity types; relationships between entity types; business rules)
 - project data model: more detailed view, matching data structure in database or data warehouse

Activity 3

- a. Data (Structured) - text, date
- b. Metadata (Property of data) - length constraint
- c. Data (Unstructured) - image
- d. Data (Unstructured) - image
- e. Metadata (Context of data)
- f. Metadata (Context of data)
- g. Metadata (Property of data) - time constraint

Activity 4

- Data are the actual details like book titles, member names, and transaction records.
- Metadata ensures the database enforces rules such as maximum length, valid formats (e.g., email), and constraints (e.g., ISBN must be unique, due date after borrow date).
- 1. System Data Identification (Essential Data):
 - a. Books
 - Book ID
 - Title
 - Author
 - ISBN
 - Publisher
 - Year of publication
 - Category/Genre
 - Availability status
 - b. Borrowers (Library Members)

- Member ID
- Name
- Address
- Phone number
- Email
- Membership type (student, faculty, external)

c. Transactions (Borrow/Return Records)

- Transaction ID
- Member ID
- Book ID
- Date borrowed
- Due date
- Date returned
- Fine (if applicable)

2. Metadata Description (Properties of the Data):

a. Books

- Book ID → Integer, Primary Key, Auto-increment
- Title → String, max 100 characters, not null
- Author → String, max 50 characters, not null
- ISBN → String, 13 characters, unique
- Publisher → String, max 50 characters
- Year of publication → Integer, must be $\leq$ current year
- Category/Genre → String, max 30 characters
- Availability status → Boolean (Available / Checked out)

b. Borrowers

- Member ID → Integer, Primary Key, Auto-increment
- Name → String, max 50 characters, not null
- Address → String, max 150 characters
- Phone number → String, 10–15 digits
- Email → String, must follow email format, unique
- Membership type → Enum ('Student', 'Faculty', 'External')

c. Transactions

- Transaction ID → Integer, Primary Key, Auto-increment
- Member ID → Integer, Foreign Key (references Borrower)
- Book ID → Integer, Foreign Key (references Book)
- Date borrowed → Date, not null
- Due date → Date, must be after borrow date
- Date returned → Date, nullable (empty if not returned)
- Fine → Decimal(6,2), default = 0